

migration, data analytics, data lake solution development, and creating real-time data pipelines with machine learning solutions. Addressing these challenges requires expertise and highly skilled resources to manage risks and mitigate them in real-time.

AWS provides engineering support through AWS Data Lab, facilitating accelerated data engineering activities through skilled AWS technical resources and lab facilities. AWS Data Lab is like a black box setup for data engineering activity to develop solutions; it comprises two categories — Build Lab and Design Lab.

Design Lab offers 1.5 to 2 days of engagement with the AWS engineering and technical team to build a solution design for data engineering projects for customers. Build Lab offers 2-5 days of intense workshop engagement with the AWS technical team to develop a working trailblazer for the full-blown architecture/solution design. There are five AWS Data Lab hubs, located in Seattle, New York, Herndon, London and Bengaluru. You can engage with these labs online or through in-person visits.

Using AWS Data Lab, you can develop solutions using AWS Quick Sight, build machine learning solutions for data analytics using SageMaker, develop data lake formation templates, and carry out real-time streaming using AWS EMR for Apache Spark. AWS Data Lab works on three principles:

- Think big to develop a robust data engineering solution
- Start small to create a prototype solution
- Scale fast to ensure all the best practices advised by the AWS technical team are incorporated in the solution design

In-memory data storage services

For Big Data processing like data aggregation or data analytics, we may need high-speed data operations like querying, updating and storing data to physical data stores. With disk-based services, this cannot be achieved to the high speed that we need, as they are physical IO operations. Modern techniques introduced later with first level and second level caching, and then data storage with solid-state devices (SSD) gave better transaction rates and fault-tolerant read-write operations.

However, these could not solve the problem and hence modern cloud architecture introduced in-memory data stores like Memcached, which can store entire databases in memory and does not need any physical IO operation. The ultra-speed design includes an in-memory data store with persistent state using a framework like Redis to handle main memory data for backend synchronisation.

Amazon MemoryDB for Redis is such an in-memory datastore with persistent facility, and is now generally available (GA) across multiple regions. It doesn't use any disk IO for non-change (read-only) operations, and uses a backend synchronised data store for data change operations like update/add/delete. This design ensures reliable, real-time synchronisation.

AWS Data Pipeline

AWS Data Pipeline is a fully managed service designed to automate the movement and transformation of data between AWS compute and storage services, as well as on-premise data sources. It helps you build reliable, managed data pipelines to collect, transform, and move your data at scale.

These are the key characteristics of this service.

Orchestrated data movement:

AWS Data Pipeline allows the orchestration of data movement between different data stores.

Data transformation: It allows you to transform data using a variety of pre-built AWS services, such as Amazon EMR and Amazon Redshift.

Dependency management: It also allows you to define dependencies between tasks, so that tasks can only run after their dependencies have completed successfully.

Error handling: AWS Data Pipeline provides several features for handling errors, such as retries and notifications.

Monitoring and alerting: It provides monitoring and alerting capabilities that allow you to track the status of your pipelines and receive notifications when there are problems.

AWS Data Pipeline works by orchestrating the movement and

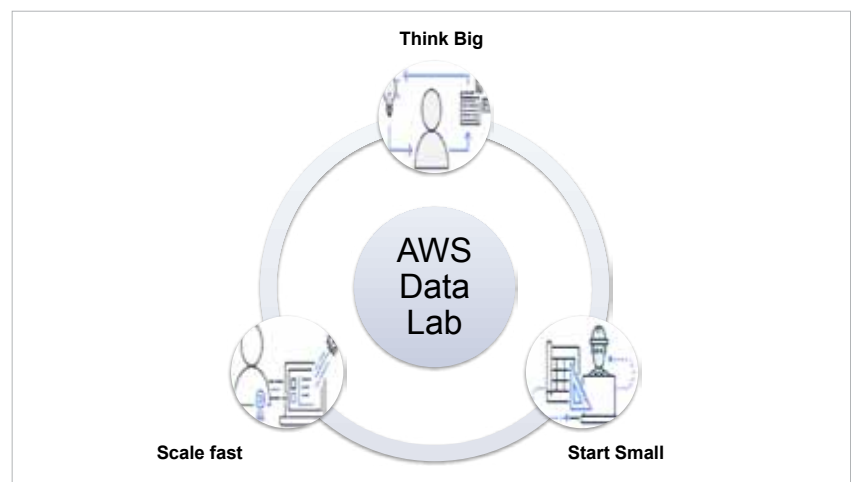


Figure 2: AWS Data Lab